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Hardware Management Project

Auxiliary Management Controller (AMC) Specification

Version 2.0

Modular Base Specification
Effective October 23, 2026

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3 Compliance with OCP Tenants

3.1 Openness

The document proposes an ecosystem-enabling set of requirements for peripherals to enable management compatibility between open systems. This allows interoperability between various device classes and host systems.

3.2 Efficiency

OEMs invest time to create specifications for industry Independent Hardware Vendors (IHVs) which must be implemented in order to support proper management by the host. IHVs invest time working with multiple OEMs to implement those requirements. The goal of the AMC specification is to standardize those various work streams into a single public OCP specification where both OEM and IHV can more effectively promulgate these requirements. Additionally, multi-vendor customer environments will benefit from the efficiencies achieved through increased device interoperability and the utilization of a common code base for system management.

3.3 Impact

This document represents a single set of OCP device manageability requirements allowing for IHV ease of development, time to market, and effective use of engineering resources. Device management ASICs could be developed allowing multiple IHVs to leverage a standardized auxiliary management controller component providing consistent management across device classes.

3.4 Scale

Large scale deployments benefit from the standardization of management capability across multiple device classes, server and device vendors which this specification provides.

Redfish, RDE and PLDM DMTF standards for management are utilized allowing for a common set of APIs and management tools regardless of hardware or software environment or size of server deployment.

3.5 Sustainability

Between customers' sustainability initiatives and demands to control energy consumption and costs, the ability to report, analyze and actuate server power usage data has become a key initiative.

The creation of a truly interoperable telemetry environment will allow businesses to datacenters, no matter the size, to more effectively meet sustainability targets. Auxiliary management controller thermal and power management capabilities can be utilized to enable this goal of minimizing power requirements and overall energy usage

4 Change Log

Date	Version #	Author	Description
7/5/2022	0.1	Chad Yoshikawa	Filled in Title, Authors, Contributors and sections 1-3
10/1/2022	0.2	Ed Tanous	Major reorganization. Rewrites to many sections
4/27/2023	0.3	Ed Tanous	Removal of old text, formatting cleanups
9/15/2023	1.0	Chad Yoshikawa	Used recent
2/12/2024	1.0	Gregg Shick	Convert to markdown
4/15/2024	1.0	John Leung	Format to work with DMTF Doc Publication tool
10/22/2026	2.0	Gregg Shick	Add Table #'s Cleanup SMC to AMC Rewrite of Security section General doc cleanup Added DMTF spec minimums Added PLDM type 7 as requirement Additional PDR type, event and repository requirements added Include DSP0283 as a *should* requirement for MCTP

5 Overview

The Auxiliary Management Controller specification defines the management interface between a host server and hardware plug-ins. For example: the host server has a baseboard management controller (BMC). The plug-in has an auxiliary management controller (AMC). Hence, the manageament interface is the interface between the BMC and the AMC.

Auxiliary Management Controllers are simple. They manage a single thermal, power and security domain. They do not manage other devices.

The host management controller and auxiliary management controller may be integrated onto the same board in the case of a tray and its SoC.

The conformance of the AMC can be validated using software tools. This enables independent hardware development and bring-up. To provide validation, this document defines a compatibility test suite (CTS) to ensure conformance to specified functional requirements.

The management interface additionally specifies SLOs (Service Level Objective) for operations such as firmware update and power management operations. These SLOs may impose constraints on the underlying hardware. For example, timely firmware update may require i3c (vs. i2c) or higher-bandwidth management links.

5.1 Architectural Example

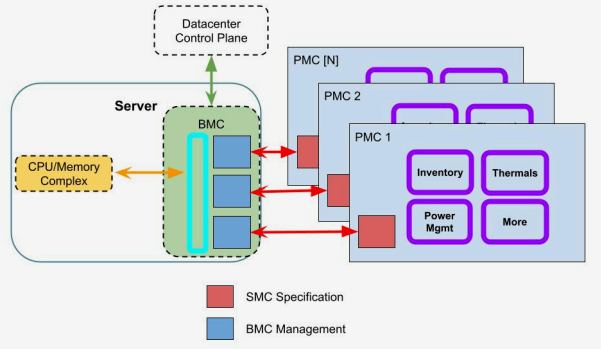


Figure 1: AMC Architectural Example Diagram

This document includes all API definitions required for managing a peripheral device from an out-of-band baseboard management controller (BMC) in the most common configuration. While other configurations may exist that this specification fulfills, the above diagram is considered the baseline. This document may reference elements of the baseline configuration as examples. Other configurations may exist.

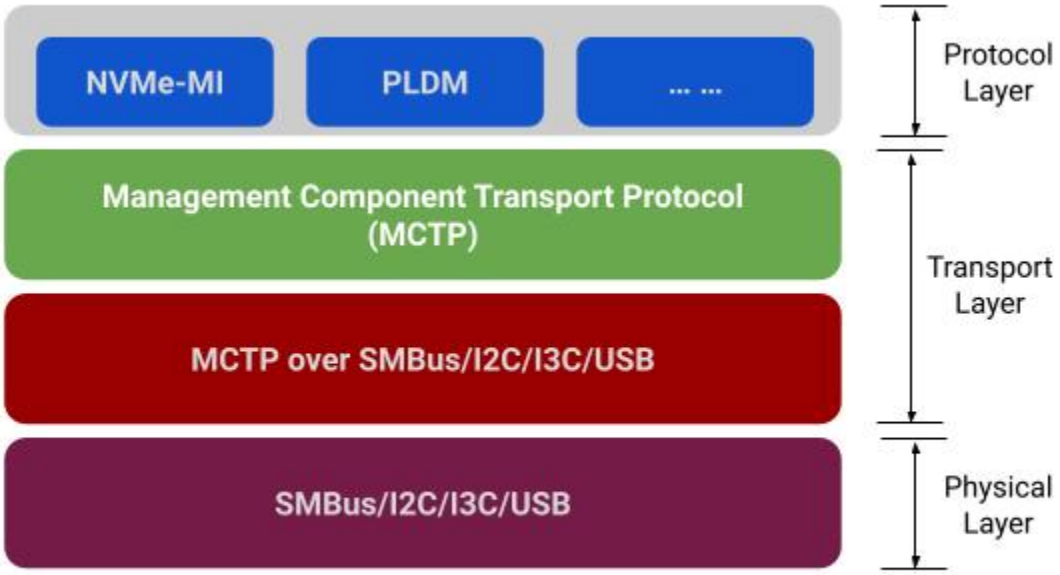
This document explicitly does not define any physical connectors, physical form factors, or specifications for non-AMC components. Other specifications (such as the Modular Hardware Management DC-SCM) may define connector specifications that may be used in deployments. For some examples, this specification may assume a PCIe 16X connector, and a single socket server with BMC.

6 Protocols

AMC devices *shall* implement DMTF DSP0236 (Management Component Transport Protocol (MCTP) Base Specification).

6.1 Physical bindings supported

Devices meeting this specification *shall* implement DMTF DSP0233 (Management Component Transport Protocol (MCTP) I3C Transport Binding Specification) OR DSP0237 (Management Component Transport Protocol (MCTP) SMBus/I2C Transport Binding Specification) OR DSP0238 (Management Component Transport Protocol (MCTP) PCIe VDM Transport Binding Specification). Additionally, DSP0283 (Management Component Transport Protocol (MCTP) Universal Serial Bus (USB) Transport Binding Specification *should* be supported. Devices *shall* implement these protocols without requiring an on-board i2c mux in the protocol path.



6.2 Protocol layers supported

6.2.1 MCTP

- Compliance with MCTP Base Protocol Specification

  i. The device *shall* comply to all endpoint requirements as specified in this section

  ii. For SMBus (Binding) Protocol, the device *shall not* be the MCTP Bus Owner but *may* act as MCTP proxy.

  iii. The device *shall* support receiving an MCTP Set Endpoint ID command at any time (Standby or Main) power is applied. For example, the BMC firmware could be updated resulting in a BMC reset. The BMC initialization sequence will go out and dynamically discover devices and assign Endpoint IDs. While in most cases the same Endpoint ID will be assigned, this is not guaranteed.

- The device *shall* comply with Section “MCTP Message Types”
- The device *shall* comply with Section “MCTP Control Protocol”
- The device *shall* comply with Section “MCTP Control Messages”
- The device *shall* comply with Section “MCTP Control Message Command Codes”, “MCTP Control Command Numbers”
- The device *shall* comply with Section “Prepare for Endpoint Discovery”
- The device *shall* comply with Section “Endpoint Discovery”
- Endpoints *should* be ready in less than 60 seconds from power ON

Table 1: MCTP Command

MCTP Control Command (DSP0236)	**Implementation**
0x01 Set Endpoint ID	Required
0x02 Get Endpoint ID	Required
0x03 Get Endpoint UUID	Required
0x04 Get MCTP Version Support	Required
0x05 Get Message Type Support	Required
0x06=Get Vendor Defined Message Support	Required
0x0B=Prepare for Endpoint Discovery	Required (PCIe VDM)
0x0C=Endpoint Discovery	Required (PCIe VDM)
0x0D=Discovery Notify	Required (PCIe VDM)

6.2.2 PLDM

Table 2: PLDM Command

PLDM Messaging Control and Discovery Command Codes (DSP0240)	**Implementation**
0x01 SetTID	Required
0x02 GetTID	Required
0x03 GetPLDMVersion	Required
0x04 GetPLDMTypes	Required
0x05 GetPLDMCommands	Required
0x06 SelectPLDMVersion	Required
0x07 NegotiateTransferParameters	Required for PLDM FW Multipart
0x08 MultipartSend	Required
0x09 MultipartReceive	Required

6.2.3 Redfish Device Enablement

- Devices *shall* conform to DSP0218 (Platform Level Data Model (PLDM) for Redfish Device Enablement)

- Devices *shall* support one RDE tree operation.
- Devices *should* support a minimum of 4 concurrent RDE tree operations.
- Devices *should* be capable of returning the requested portions of the Redfish tree in less than 2 seconds.

Table 3: RDE Command

RDE Command		Implementation
0x01 NegotiateRedfishParameters		Required
	DeviceCapabilitiesFlags	Support atomic resource read to enable consistent reads
	DeviceFeatureSupport	Head and replace optional. Read mandatory. Other features are optional for inventory-only devices and mandatory for full support.
	DeviceConfigurationSignature	Required as documented
0x02 NegotiateMediumParameters		Required
0x03 GetSchemaDictionary		Required
0x04 GetSchemaURI		Required
0x05 GetResourceETag		Required
0x08 GetRegistryCount		Required for full event support
0x09 GetRegistryDetails		Required for full event support
0x0A SelectRegistryVersion		Required for full event support
0x0B GetMessageRegistry		Required for full event support
0x10 RDEOperationInit		Required for full event support
	0=OPERATION_HEAD	Required
	1=OPERATION_READ	Required
	2=OPERATION_CREATE	Conditional - Required for write support
	3=OPERATION_DELETE	Conditional - Required for write supportired
	4=OPERATION_UPDATE	Conditional - Required for write support
	6=OPRERATION_ACTION	Conditional - Required for write support
0x11 SupplyCustomRequestParameters		Required
0x12 RetrieveCustomResponseParameters		Required
0x13 RDEOperationComplete		Required
0x14 RDEOperationStatus		Required
0x15 RDEOperationKill		Required
0x16 RDEOperationEnumerate		Required
0x30 RDEMultipartSend		Required for write support
0x31 RDEMultipartReceive		Required

6.3 Device Identification

Devices *shall* expose an interface to a Platform Management FRU Information Storage Definition-compatible API. The document does not specify where this FRU payload must be physically implemented within the card. Implementations may choose to implement in a physical eeprom device, or a virtual eeprom device. Care should be taken in platform design to ensure that the eeprom is available in the required power states consistent with the needs of the baseline server, but this specification does not specify which power states it will be available in.

The FRU information *shall* contain the following fields. Other fields may be populated.

- Manufacturer Name
- Product Name
- Product Serial Number

6.4 Device Classes

The following device classes *shall* be supported: Accelerator, NIC, DPU (SmartNIC), Storage Tray, Memory (CXL).

7 API Surface

Table 4 specifies the required support for various device classes. “R” signifies that support is required. “C” signifies that support is required if the feature exists. For example, an Accelerator device may not have a Fan and thus Fan Control is conditional.

Table 4: API Surface

Subsystem	Thermal	Inventory	Software Management	Fan Control	Security	Power Management
Accelerator	R	R	R	C	R	C
NICs	C	R	R	C	R	C
DPUs	R	R	R	C	R	C
Storage Tray	C	R	R	C	R	C
Memory (CXL)	R	R	C	C	R	C

7.1 Thermal management subsystem

The thermal management subsystem is intended to allow AMC devices to be managed by a closed loop thermal system. Implementation of the thermal management subsystem requires support for the following Redfish resources and properties.

Sensor

The SensorCollection resource *shall* implement one or more sensor resources representing the thermal temperature of the device.

The sensor *shall* have the following properties supported.

- **ReadingUnits** *shall* be presented and implement as “Cel”
- **ReadingType** *shall* be present and implemented as “Temperature”
- Threshold properties of **UpperCritical**, **UpperFatal**, **LowerCritical**, and **LowerFatal** *shall* be implemented and represent the design limits of the AMC device in question. Devices with unlimited thermal design limits *shall* omit these properties.

If the AMC device possess fans, the AMC SensorCollection shall implement a Sensor with the following properties:

- **ReadingUnits** *shall* be present and implemented as “{rev}/min”
- **ReadingType** *shall* be present and implemented as “Rotational”

Devices implementing a single replaceable component may implement PLDM type 2 for sensor readings.

PLDM Type 2 Sensor Requirements

Table 5 specifies the support requirement for PLDM for Platform Monitoring and Control (DSP0248)

Table 5: PLDM Type 2 Command

Command Codes		**Implementation**
Terminus Command		
	0x04 SetEventReceiver	Required for RDE Alert
	0x05 GetEventReceiver	Required for RDE Alert
	0x0B PollForPlatformEventMessage	Required for RDE Alert
	0x0C EventMessageSupported	Required
	0x0D EventMessageBufferSize	Required for RDE Alert if MSG size > 256 bytes
Numeric Sensor Commands		
	0x11 GetSensorReading	Required
	0x12 GenSensorThreshold	Required
	0x15 GetSensorHysteresis	Required
State Sensor Commands		
	0x21 GetStateSensorReadings	Required
PDR Repository Commands		
	0x50 GetPDRRepositoryInfo	Required
	0x51 GetPDR	Required
	0x53 GetPDRRepositorySignature	Required for RDE
PLDM Event Types		
	0x02 redfishTaskExecutedEvent	Required if implementation cannot complete commands quickly enough to avoid spawning RDE tasks
	0x03 redfishMessageEvent	Required for redfish eventing
	0x04 pldmPDRRepositoryChgEvent	Required
	0x05 pldmMessagePollEvent	Required
	0x51 GetPDR	Required for RDE
PDR Type Values		
	2 = Numeric Sensor PDR	Required
	4= State Sensor PDR	Required
	22 = Redfish Resource PDR	Required for RDE
	23 = Redfish Entity Association PDR	Required
	24 = Redfish Action PDR	Required
	126 = OEM Device PDR	Required

Devices implementing multiple replaceable components *shall* implement ThermalSubsystem

over RDE.

ThermalSubsystem

An AMC Redfish ThermalSubsystem *shall* be implemented, with the following properties:

- *Fans*: With Fan resources representing the fans physically present on this device.
- Fan resources *shall* contain the following properties
  - *SpeedPercent*

7.2 Inventory Management

Requirements within this section are intended to allow inventory management and control of a given device. AMC devices *shall* implement:

ChassisCollection

The ChassisCollection in the device *shall* contain one or more Chassis Resources representing the device in question. Chassis resources shall implement the following properties:

- Model: The value of this property *shall* match the “Product Name” field present in the FRU identification from section 5.3
- Manufacturer:The value of this property *shall* match the “Manufacturer Name” field present in the FRU identification from section 5.3
- SerialNumber: The value of this property *shall* match the “Serial Number” field present in the FRU identification from section 5.3

AMC devices *may* implement more than one Chassis resource, for representing physical subsystems within the device. Within the AMC chassis collection, there *shall* be only one Chassis instance (referred to further as the “root”) that does not possess a ContainedBy attribute, and is intended to represent the overall containment of the device. All other devices *shall* have a ContainedBy Link, traceable to the root device. Root devices *shall* implement a “Contains” property representing the devices containment

7.3 File I/O

AMC devices *shall* conform to DSP0242 PLDM for File Transfer to support file I/O capability necessary to provide device logging information.

7.4 Firmware and Software Update

AMC devices may have one or multiple updateable firmware or software components.

7.4.1 Single Update

AMC devices with a single updateable firmware or software component *shall* implement PLDM for Firmware Update (type 5).

Table 6: PLDM Type 5 Command

PLDM for Firmware Update	Implementation
0x01 QueryDeviceIdentifiers	Required
0x02 GetFirmwareParameters	Required
0x03 QueryDownstreamDevices	Required
0x04 QueryDowstreamIdentifiers	Required
0x05 GetDownstreamFirmwareParameters	Required
0x10 RequestUpdate	Required
0x11 GetPackageData	Required
0x13 PassComponentTable	Required
0x14 UpdateComponent	Required
0x15 RequestFirmwareData	Required
0x16 TransferComplete	Required
0x17 VerifyComplete	Required
0x18 ApplyComplete	Required
0x19 GetMetaData	Required
0x1A ActivateFirmware	Required
0x1B GetStatus	Required
0x1B GetStatus	Required
0x1C CancelUpdateComponent	Required
0x1D CancelUpdate	Required
0x1E ActivatePendingComponentImageSet	Required
0x1F ActivatePendingComponentImage	Required
0x20 RequestDownstreamDeviceUpdate	Required

7.4.2 Multiple Update

AMC devices with multiple updateable firmware or software components *shall* support an UpdateService resource. The UpdateService resource *shall* implement one or more of FirmwareInventory, or SoftwareInventory collections. Collections shall contain at least one member of type SoftwareInventory, implementing the following properties

- **Version**: The software version running on the AMC device.
- **Updateable**: Whether or not the device supports update. For AMC devices, this shall be set to True.
- **SoftwareID**: Unique identifier for this devices firmware type.
- **AdditionalVersions**: Devices that fit the Redfish descriptions in these properties *shall* implement AdditionalVersions, with the appropriate subproperties. Devices that do not fit the description *shall* omit the AdditionalVersions Property.

AMC devices *shall* be updated in 1 minute or less, measured in the time that the device is unavailable, and 5 minutes or less from the time the update is requested, including all data transfers to the device.

7.5 Fan Control

AMC devices containing fans *shall* implement control and monitoring of those fans through the RDE interface. Devices *shall* support the Redfish Control schema for fan control within a system. AMC devices may run internal control loops in addition to the control loops presented on the RDE interface.

7.6 Power Management

If the AMC device supports reset, the Redfish Chassis.Reset action *shall* be supported,

If an AMC device provides power metrics, the EnvironmentMetric resource and Sensor collection resource *shall* implement the following properties where supported:

- EnergykWh or EnergyJoules
- PowerWatts
- PowerLimitWatts
- ResetMetrics
- AverageReading
- AveragingInterval

7.7 Security

Requirements within this section are intended to define security and SPDM requirements for AMC controllers applicable across supported device classes. Note - SPDM Requirements adopted from the OCP Datacenter NVMe® SSD Specification version 2.6.

7.7.1 SPDM

- *Shall* support SPDM 1.2.2 (DSP0274) or later while maintaining backward compatibility with SPDM 1.1. Signature endianness *shall* follow SPDM 1.2.2 or later.
- *Shall* comply with the revised PCI-SIG Component Measurement and Authentication (CMA-SPDM) ECN that was published on September 20, 2023.
- *Shall* comply with the Secured Messages using SPDM over MCTP Binding Specification (DSP0276), Version 1.1.1 or later.
- *Shall* comply with the Secured Messages using SPDM Specification (DSP0277), Version 1.1.1 or later. Mutual authentication is not required.
- *Shall* comply with the Security Protocol and Data Model (SPDM) over MCTP Binding Specification (DSP0275), Version 1.0.2 or later.
- Must not drop the response if RESPOND_IF_READY is not received within WTMax amount of time after sending an ERROR message with an ErrorCode equal to ResponseNotReady.
- *Shall* be ready to respond to SPDM requests within a maximum of 2 second after main power is applied.
- SPDM MCTP interpacket delay *shall* be less than 40 ms.
- *Shall* support a DataTransferSize of at least 4 KiB.
- If not ready to accept a new request message, the AMC *shall* respond with an ERROR response message with an ErrorCode of Busy (3h) (i.e., the device shall not silently discard the request message).
- If a request is received out of order, the AMC *shall* respond with an ERROR response message with an ErrorCode of RequestResynch (i.e., 43h) for that request and for all

subsequent requests until a GET_VERSION command is received and processed. The device *shall not* silently discard requests due to an out of order request.

Table 7 specifies AMC response code requirements for SPDM.

Table 7: SPDM Response

SPDM Response	Implementation	Notes
0x01 DIGESTS	Required	
0x02 CERTIFICATE	Required	
0x03 CHALLENGE_AUTH	Required	Responder shall be able to provide the measurement summary type requested
		0x0 - No measurement summary hash requested
		0x1 - TCB measurements only
		0xFF - All measurements
0x04 VERSION	Required	
0x05 CHUNK_SEND_ACK	Optional	
0x06 CHUNK_RESPONSE	Optional	
0x60 MEASUREMENTS	Required	
0x61 CAPABILITIES	Required	CERT_CAP - Responder shall support DIGESTS and CERTIFICATE response messages.
		CHAL_CAP - Responder shall support CHALLENGE_AUTH response messages.
		MEAS_CAP - Responder shall support MEASUREMENTS response messages and can generate signatures - MEAS_CAP = 10b
0x63 ALGORITHMS	Required	**Implementation Guidance**
		BaseAsymSel
		TPM_ALG_RSASSA_3072 [CMA, CNSA, OCP] Allowed
		TPM_ALG_ECDSA_ECC_NIST_P256 [CMA] Allowed
		TPM_ALG_ECDSA_ECC_NIST_P384 [CMA, CNSA, OCP] Preferred
		BaseHashSel
		TPM_ALG_SHA_256 [CMA] Allowed
		TPM_ALG_SHA_384 [CMA, CNSA, OCP] Preferred
		MeasurementHashAlgo
		TPM_ALG_SHA_256 [CMA] Allowed
		TPM_ALG_SHA_384 [CMA, CNSA, OCP] Preferred
0x64 KEY_EXCHANGE_RSP	Required	
0x65 FINISH_RSP	Required	
0x66 PSK_EXCHANGE_RSP	Optional	
0x67 PSK_FINISH_RSP	Optional	
0x68 HEARTBEAT_ACK	Optional	
0x6C END_SESSION_ACK	Required	

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SPDM Response	Implementation	Notes
0x6D CSR	Recommended	
0x6E SET_CERTIFICATE_RSP	Recommended	
0x7E VENDOR_DEFINED_RESPONSE	Optional	
0x7F ERROR	Required	

7.7.2 SPDM Certificate Requirements

- The Root CA Trusted Certificate Authority *shall* be the device vendor and shall be the same across all device models developed by the device vendor.
- Self-signed certificates are prohibited.
- The AMC *shall* support the AliasCert or DeviceCert model in slot 0.

7.7.3 SPDM Firmware Measurements

- The AMC *shall* be able to generate signed measurements (i.e., MEAS_CAP field shall be set to 10b) and *shall* support the following Measurement block types:

Table 8: SPDM Measurements

DMTFSpecMeasurementValueType	Requirement
[7] - 0b: Digest, 1b: Raw Bit Stream	Required
[6:0] = 00h: immutable ROM	Required
[6:0] = 01h: mutable firmware	Required
[6:0] = 02h: hardware configuration, such as straps, debug modes	Recommended
[6:0] = 03h: firmware configuration (e.g., configurable firmware policy)	Recommended
[6:0] = 04h: Measurement manifest	Required
[6:0] = 05h: Structured representation of debug and device mode	Optional
[6:0] = 06h: Mutable firmware’s version number	Optional
[6:0] = 07h: Mutable firmware’s security version number, which should be formatted as an 8-byte unsigned integer	Optional

- The AMC *shall* always return measurements of the most recently activated firmware and current configuration without requiring a power cycle or any type of PCIe reset. Measurements of the activated firmware image are acceptable.
- The AMC *shall* report the same measurements blocks and values for all devices in a product family (e.g., devices with the same model but different capacity points, endurance, or form factor).
- The AMC *shall* provide a reference integrity measurement manifest (RIM) for the firmware modules and configuration settings for components using the CoRIM schema as defined by IETF (refer to the latest version of draft-ietf-rats-corim - Concise Reference Integrity Manifest or the RFC (if any) published from that Internet-Draft).

- The reference measurements in the CoRIM schema may be expressed using the CoMID tags and/or the CoSWID tags. If the CoRIM has reference measurements from sub-components from different manufacturers, then the reference values of the sub-components *shall* be captured in CoSWID tags with their individual signatures.
- The RIM *shall* apply to all devices in a product family (e.g., devices with the same model but different capacity points, endurance, or form factor). The RIM *shall not* include any instance specific information (e.g., any device specific identifiers such as Serial Number, UUID, etc.).
- The RIM *shall* be signed by the device manufacturer using the COSE schema (refer to RFC 8152 - CBOR Object Signing and Encryption (COSE) (ietf.org)). The device manufacturer *shall* use an approved digital signature algorithm (see SEC-48).
- The device manufacturer *shall* provide the customer with a digital certificate with the public key that is able to be used to verify the RIM signatures. This digital certificate *shall* validate (e.g., via direct signature or certificate chain) to the Root CA Trusted Certificate Authority
- The RIM *shall* contain the hash values of each firmware module using an approved hashing algorithm (see SEC-48). ASCII text string values in the RIMs are acceptable if the string length is less than 128 bytes and the string is descriptive (e.g., a module identifier string, a product identifier string, or firmware revision string).

8 Glossary of Terms

This section provides definitions for terms used in this document.

Server. Machine hardware that contains a Auxiliary plug-in. Auxiliary containers are typically Servers but are not required to be so. So we use the more general term Host for a Auxiliary container.

Host. A generalization of a Auxiliary container that includes Servers (for PCIe Plug-In Auxiliarys) and motherboards (for SoC Auxiliarys). A Host is managed by a logical Host Management Controller (HMC).

AMC Devices. A dependent group of hardware that is managed by a logical Auxiliary Management Controller (AMC). AMC Devices typically are smaller than their Host, are terminal points in the management graph, and contain a single power & thermal domain.

AMC. Auxiliary Management Controller provides a management API to the AMC device. AMCs may be backed by one or more discrete hardware components. AMC is typically a low-cost microcontroller running a RTOS with no external DRAM.

Terminal Hardware. Hardware that is an endpoint in the management graph. In other words, terminal hardware does not itself manage other hardware. AMC devices are terminal hardware.

9 References

Implementations shall support the following minimum standard revisions

Specification	Title	Revision
DSP0218	Platform Level Data Model (PLDM) for Redfish Device Enablement	1.2.0 and later
DSP0233	Management Component Transport Protocol (MCTP) I3C Transport Binding Specification	1.0.1 and later
DSP0236	Management Component Transport Protocol (MCTP) Base Specification	1.3.3 and later
DSP0237	Management Component Transport Protocol (MCTP) SMBus/I2C Transport Binding Specification	1.2.0 and later
DSP0238	Management Component Transport Protocol (MCTP) PCIe VDM Transport Binding Specification	1.4.0 and later
DSP0283	Management Component Transport Protocol (MCTP) Universal Serial Bus (USB) Transport Binding Specification	1.1.1 and later
DSP0240	Platform Level Data Model (PLDM) Base Specification	1.2.0 and later
DSP0241	Platform Level Data Model (PLDM) Over MCTP Binding Specification	1.0.0 and later
DSP0248	Platform Level Data Model (PLDM) for Platform Monitoring and Control Specification	1.3.0 and later
DSP0267	Platform Level Data Model (PLDM) for Firmware Update Specification	1.3.0 and later
DSP0274	Security Protocol and Data Model (SPDM) Specification	1.2.2 and later with backward compatability to 1.1
DSP0242	Platform Level Data Model (PLDM) for File Transfer Specification	1.0.1
	OCP Datacenter NVMe® SSD Specification	2.6